MATHS MANTHAN

• TUITION CLASSES •

e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 06 (SOLUTIONS)
Chapter 3: Pair of Linear Equations in Two Variables

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | (3 × 1 = 3 Marks)

Q1. The area of the triangle formed by the line $\frac{x}{a}+\frac{y}{b}=1$ with the coordinate axes is: [1]

(A) ab (B) $2ab$ (C) $\frac{1}{2}ab$ (D) $\frac{1}{4}ab$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) $\frac{1}{2}ab$** [0.5 Mark]

Justification: Setting $y=0 \implies x=a$ and $x=0 \implies y=b$ gives intercepts $A(a,0)$ and $B(0,b)$ with origin $O(0,0)$. [0.5 Mark]

Right $\triangle OAB$ has area $= \frac{1}{2}\times OA\times OB = \mathbf{\frac{1}{2}ab}$ (for $a,b>0$).

Q2. If $37x+43y=123$ and $43x+37y=117$, then the value of $x+y$ is: [1]

(A) 1 (B) 2 (C) 3 (D) 4

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 3** [0.5 Mark]

Justification: Adding both equations: $(37+43)x+(43+37)y=123+117 \implies 80(x+y)=240 \implies x+y=\frac{240}{80}=\mathbf{3}$. [0.5 Mark]

Q3. In the above problem ($37x+43y=123$ and $43x+37y=117$), the value of $x-y$ is: [1]

(A) 1 (B) -1 (C) 2 (D) -2

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) -1** [0.5 Mark]

Justification: Subtracting the second equation from the first: [0.5 Mark]

$(37-43)x+(43-37)y=123-117 \implies -6x+6y=6 \implies -6(x-y)=6 \implies x-y=\mathbf{-1}$.

SECTION B: Assertion-Reason Question — MATHS MANTHAN | (1 × 1 = 1 Mark)

Q4. **Assertion (A):** If the system of equations $3x + 6y = 10$ and $2x - ky + 5 = 0$ is inconsistent, then $k = -4$.

Reason (R): The system $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is inconsistent if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A)** [0.5 Mark]

Verification: Rewriting first: $3x + 6y - 10 = 0$. Comparing ratios: $\frac{3}{2} = \frac{6}{-k} \implies -3k = 12 \implies k = -4$, and $\frac{-10}{5} = -2 \neq \frac{3}{2}$. [0.5 Mark]

Hence both (A) and (R) are true and (R) is the correct explanation of (A).

SECTION C: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q5. Solve the following pair of linear equations by substitution:

$$s - t = 3 \quad \text{and} \quad \frac{s}{3} + \frac{t}{2} = 6$$

[2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Clearing Fractions & Substituting $s = t + 3$ [1 Mark]

From $s - t = 3$, we have $s = t + 3$ — (i). Taking LCM 6 in second equation: $2s + 3t = 36$ — (ii).

Substituting (i) into (ii): $2(t + 3) + 3t = 36 \implies 2t + 6 + 3t = 36 \implies 5t = 30 \implies t = 6$.

Step 2: Finding s [1 Mark]

Putting $t = 6$ in (i): $s = 6 + 3 = 9$.

$$\boxed{\mathbf{s = 9, \quad t = 6}}$$

Q6. If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, determine the value of k. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Writing in Standard Form & Applying Condition [1 Mark]

$3x + 2ky - 2 = 0$ and $2x + 5y + 1 = 0$. For parallel lines, $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$:

Step 2: Solving for k [1 Mark]

$$\frac{3}{2} = \frac{2k}{5} \implies 4k = 15 \implies \boxed{\mathbf{k = \frac{15}{4}}} \qquad \left(\text{and } \frac{3}{2} \neq \frac{-2}{1} \text{ holds}\right)$$

Q7. Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as old as Sonu. How old are Nuri and Sonu? [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let Nuri's present age be x years and Sonu's present age be y years.

Step 1: Setting Up Age Equations [1.5 Marks]

5 years ago: $x - 5 = 3(y - 5) \implies x - 5 = 3y - 15 \implies x - 3y = -10$ — (i)

10 years later: $x + 10 = 2(y + 10) \implies x + 10 = 2y + 20 \implies x - 2y = 10$ — (ii)

Step 2: Solving by Elimination [1.5 Marks]

Subtracting (i) from (ii): $(x - 2y) - (x - 3y) = 10 - (-10) \implies y = 20$ years.

Putting $y = 20$ in (ii): $x - 2(20) = 10 \implies x = 50$ years.

Nuri = **50** years, Sonu = **20** years

Q8. A chemist has one solution which is 50% acid and another solution which is 25% acid. How much of each should be mixed to make 10 litres of a 40% acid solution? [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let volume of 50% acid solution be x litres and of 25% acid solution be y litres.

Formula / Theorem Used: Pure Acid Content = (Concentration %) × (Solution Volume)

Step 1: Total Volume & Pure Acid Equations [2 Marks]

Total mixture volume: $x + y = 10$ — (i)

Pure acid balance: $\frac{50}{100}x + \frac{25}{100}y = \frac{40}{100} \times 10 \implies \frac{x}{2} + \frac{y}{4} = 4.$

Multiplying through by 4: $2x + y = 16$ — (ii)

Step 2: Solving for x and y [2 Marks]

Subtracting (i) from (ii): $(2x + y) - (x + y) = 16 - 10 \implies x = 6$ litres.

From (i): $6 + y = 10 \implies y = 4$ litres.

Step 3: Conclusion [1 Mark]

6 litres of 50% acid solution and **4** litres of 25% acid solution

Q9. CASE STUDY: Eco-Tourism Boating Expedition & River Dynamics

An eco-tourism enterprise runs solar-powered safari boats on a river. The speed of the boat in still water is x km/h and the speed of the river current is y km/h. The downstream speed is $(x+y)$ km/h and the upstream speed is $(x-y)$ km/h. During a test trial, the boat covers 21 km downstream in 3 hours, and on the return trip upstream, it covers 12 km in 4 hours. [4]

(i) Formulate the pair of linear equations in x and y representing the downstream and upstream journeys. [1]

(ii) Find the speed of the safari boat in still water (x). [1]

(iii) Find the speed of the river current (y). [1]

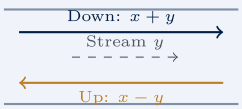
(iv) What total time will the boat take to travel 10 km downstream and then return back 10 km upstream? [1]

— OR —

Find the ratio of downstream speed to upstream speed. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Scenario Data: Downstream distance = 21 km in 3 h; upstream distance = 12 km in 4 h.



Formula / Theorem Used: $\text{Speed} = \dfrac{\text{Distance}}{\text{Time}}$; $v_{\text{down}} = x+y$, $v_{\text{up}} = x-y$

Sub-part (i): Linear Equations for Downstream & Upstream [1 Mark]

$x+y = \frac{21}{3} \implies x+y = 7$ — (i) and $x-y = \frac{12}{4} \implies x-y = 3$ — (ii)

Sub-part (ii): Speed of Safari Boat in Still Water (x) [1 Mark]

Adding (i) and (ii): $2x = 10 \implies \boxed{\mathbf{x = 5}\ \text{km/h}}$.

Sub-part (iii): Speed of River Current (y) [1 Mark]

From (i): $5+y = 7 \implies \boxed{\mathbf{y = 2}\ \text{km/h}}$.

Sub-part (iv): Total Time for 10 km Downstream + 10 km Upstream [1 Mark]

Downstream speed = 7 km/h, Upstream speed = 3 km/h:

$$\text{Total Time} = \frac{10}{7} + \frac{10}{3} = \frac{30+70}{21} = \boxed{\mathbf{\frac{100}{21}}\ \text{hours}\ \left(= \mathbf{4\frac{16}{21}}\ \text{h}\right)}$$

Sub-part (iv) [OR Option]: Ratio of Downstream to Upstream Speed [1 Mark]

Ratio $= (x+y) : (x-y) = 7 : 3 \implies \boxed{\mathbf{7 : 3}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★